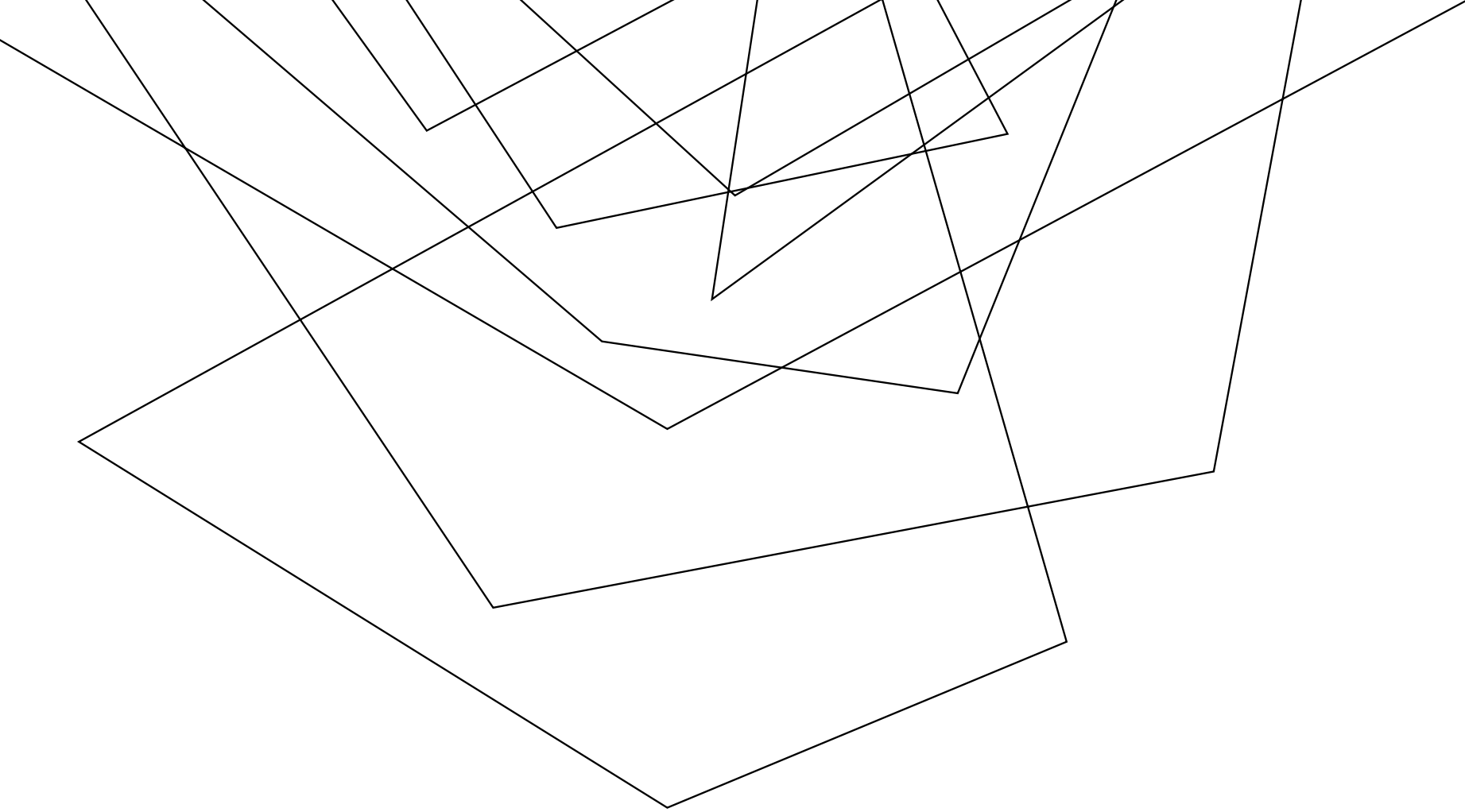




TECHNICAL ASSESSMENT

NIKKI WATSON

AGENDA

Processing

Question 1

Question 2

Question 3

Further Exploration

An abstract graphic design featuring two thin, dark gray lines that intersect on a light gray background. One line runs diagonally from the top-left towards the bottom-right, while the other runs from the top-right towards the bottom-left. The intersection point is located to the left of the text.

DATA PREPROCESSING

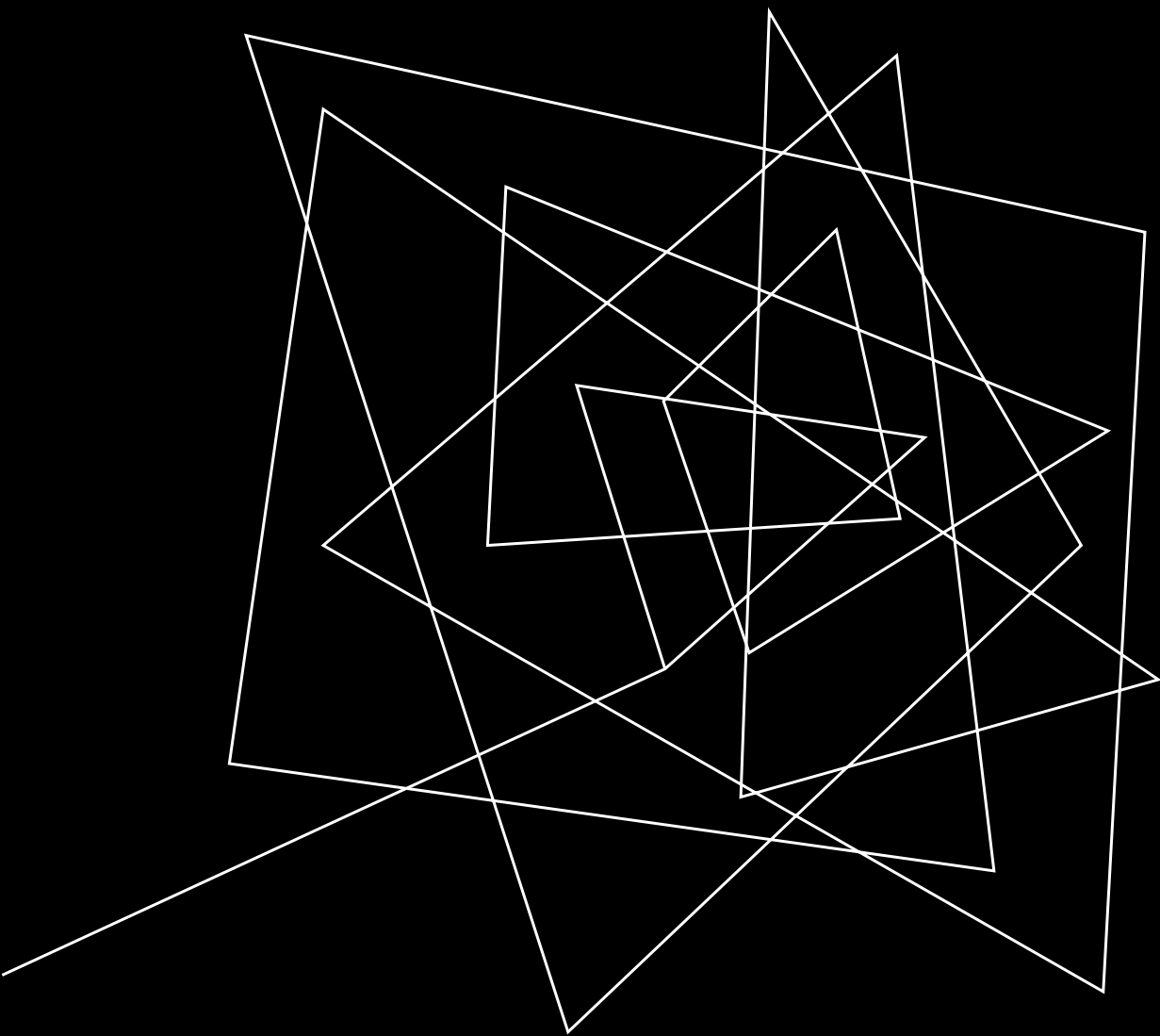
PREPROCESSING STEPS

Public Records Dataset:

1. Transform date columns into datetime format
2. Drop all-null columns (plaintiff_name, attorney_name, docket_number)
3. Examine outliers or default data-entry values

Loan Performance Dataset:

- This dataset is usable as-is and no additional processing was needed



QUESTION 1

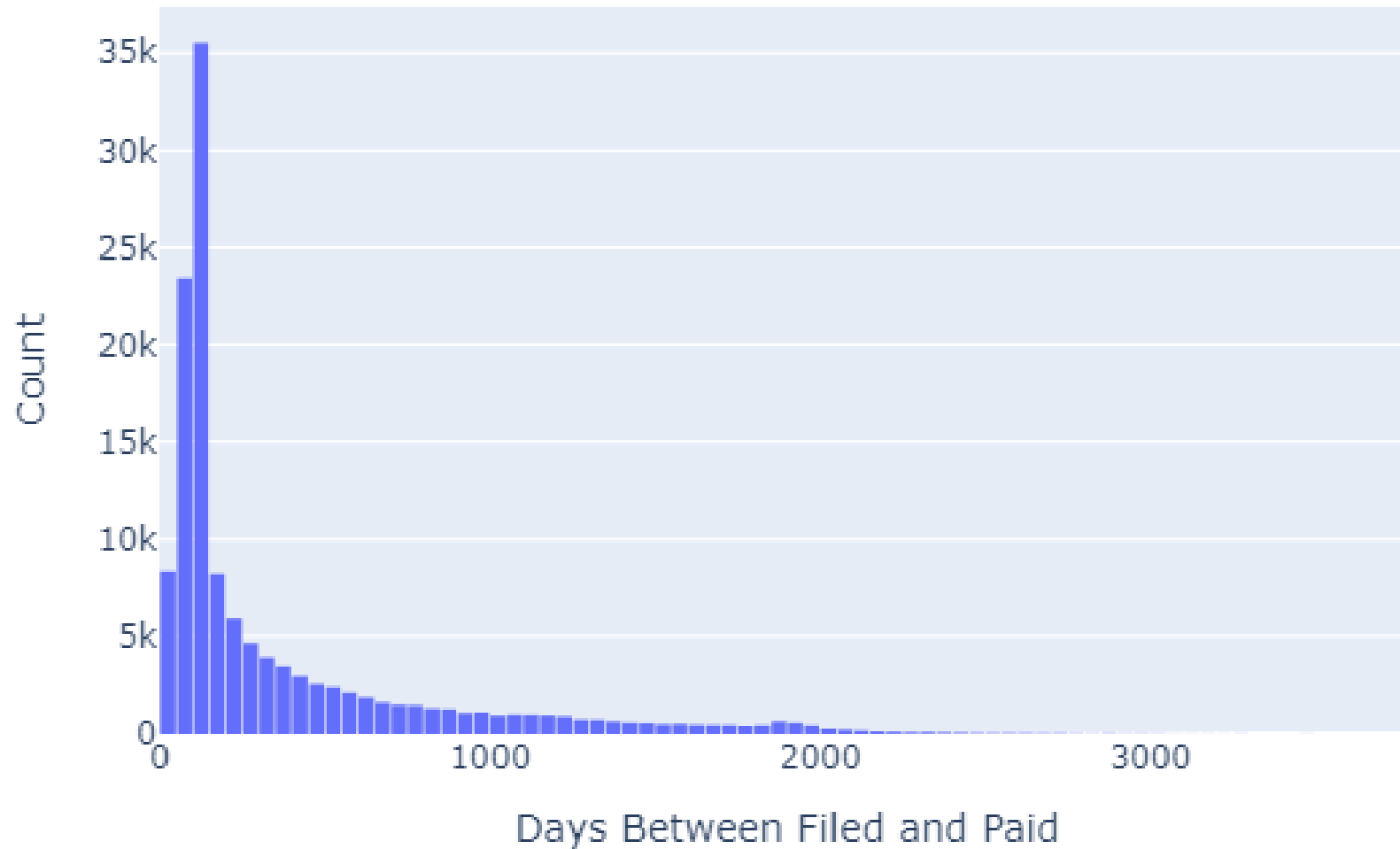
REGARDING THE PUBLIC RECORDS DATA,
ARE THERE ANY ANOMALIES? WHAT ARE
THEY AND HOW DID YOU FIND THEM?



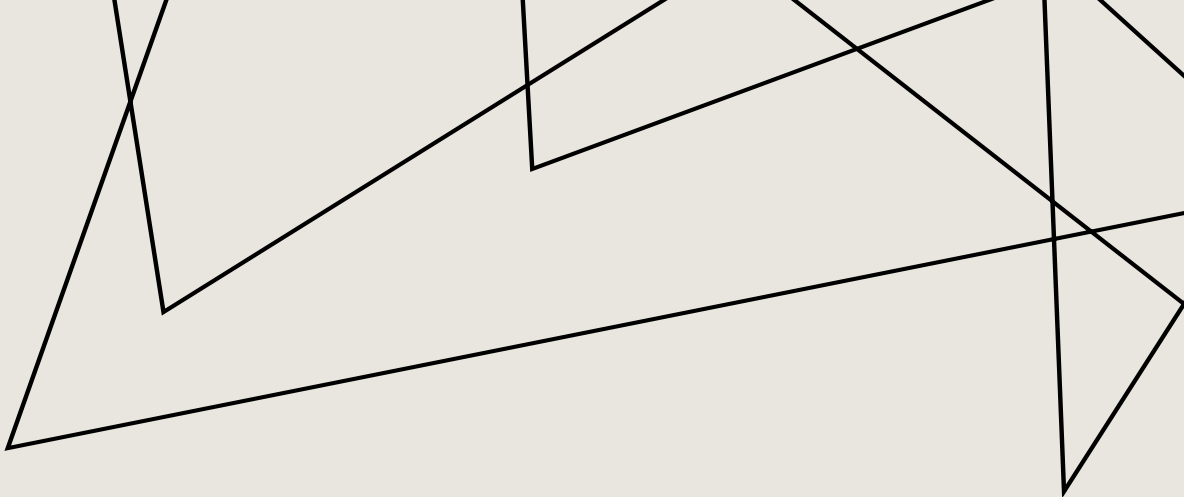
APPROACH

1. Examine the *Amount* variable
 - Z score
2. Examine the categorical variables for the *Public Record Type* and *Public Record Source Type*
 - Normalized comparison
3. Examine dates and lag time using the *Filed Date*, *Verified Date*, and *Paid Date* variables
 - Distribution analysis

DISTRIBUTION OF LOAN DURATION



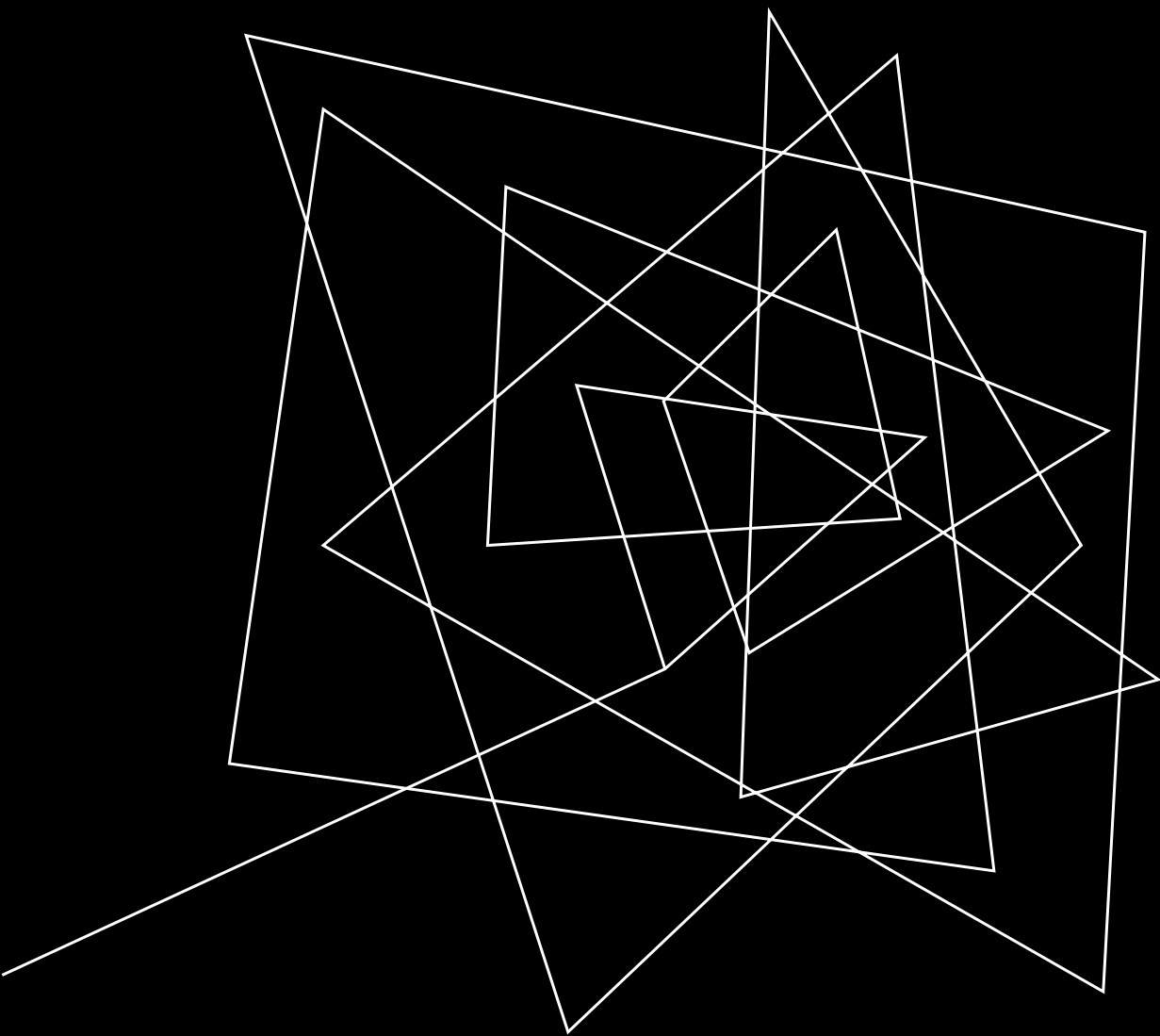
Outliers seem to include records that were paid at least 2000 days after they were filed. This accounts for less than 0.01% of the total data.



SUMMARY

Plausible constraints for anomalies include the following:

- An amount greater than \$110,338
- A public record type of FT (Federal Tax Lien)
- A public record source of either RD (Recorder of Deeds), CC (County Clerk), or CR (County Recorder)
- Records that were paid at least 2000 days or more later than they were filed



QUESTION 2

REGARDING THE PUBLIC RECORDS DATA,
PICK A VARIABLE TO INVESTIGATE AND
EVALUATE THE DISTRIBUTION – IS THERE
ANYTHING INTERESTING?



APPROACH

1. Examine the kind of business

- *Justification:* I believed that it could reveal information about what industries are most represented which could shed light on industries to take notice of both for risk assessment but also for high-priority clientele.
- There were only 2 unique identifiers in this column, so I shifted focus to a more impactful variable

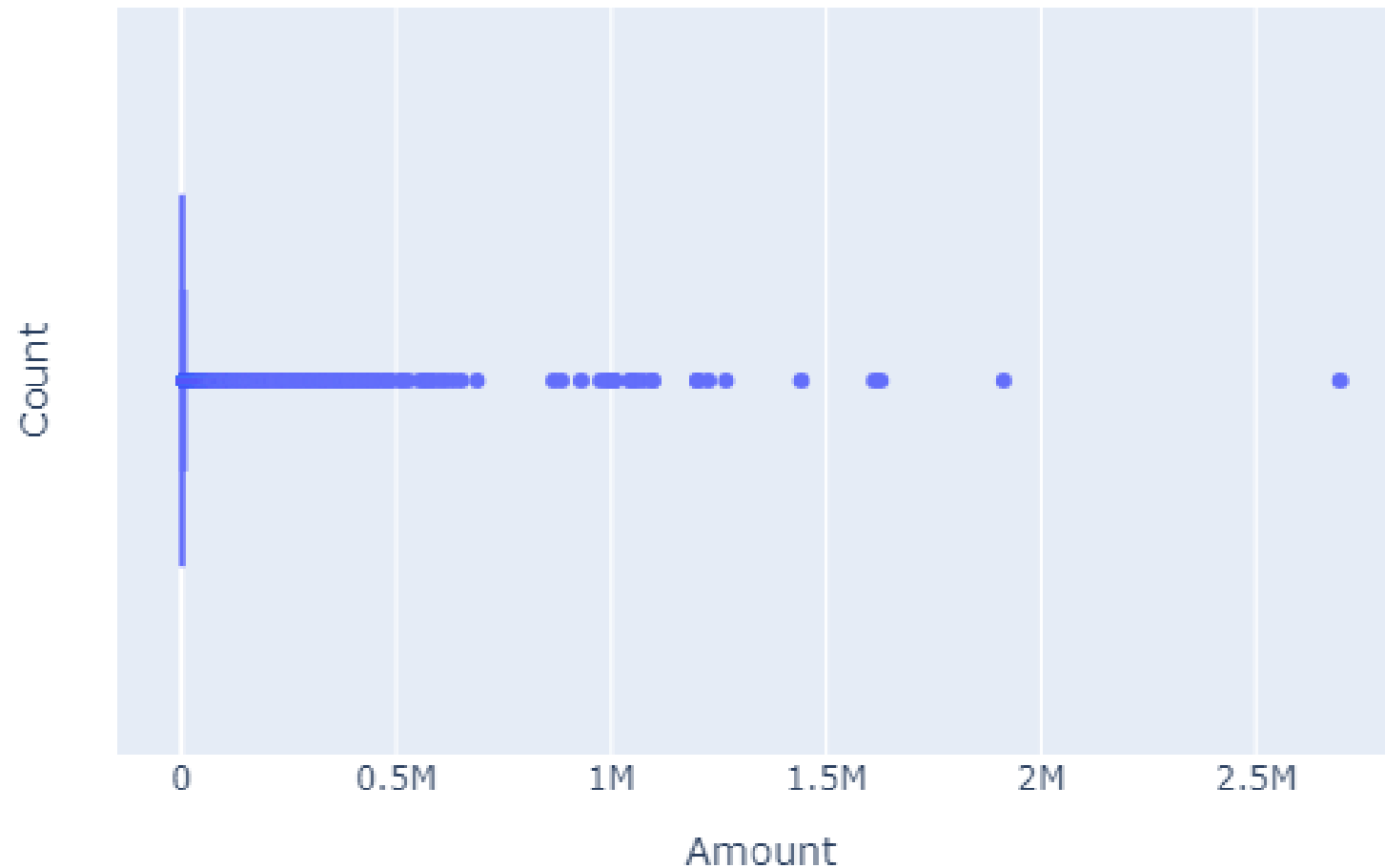
2. Examine the amount of the loan

- *Justification:* It would be worthwhile to see key statistics and analysis about the loan amounts and could contribute to the identification of key focus areas

CONSIDERATIONS

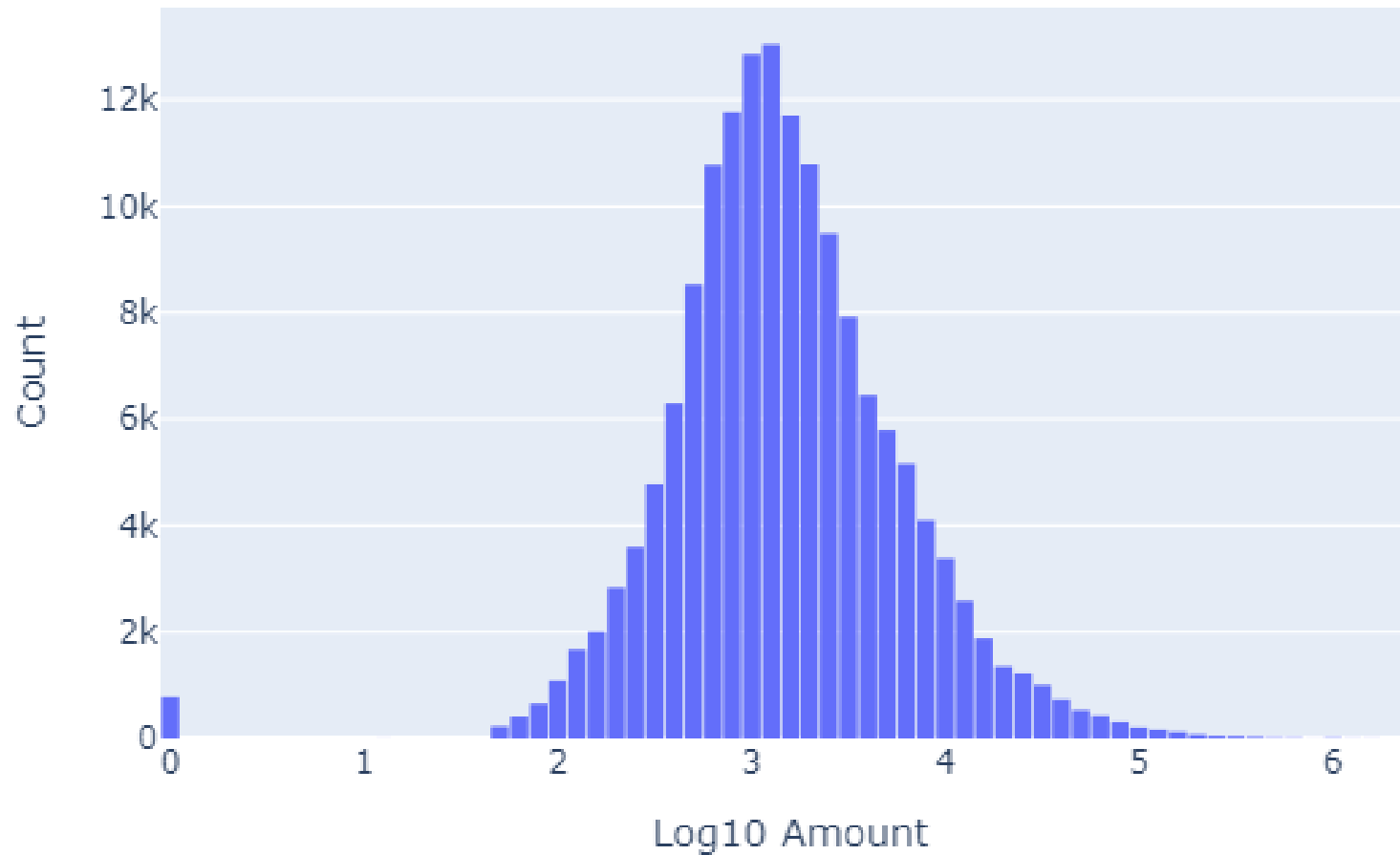
- There are 2 records considered an outlier that I believe to be entries that represent a default value (9999999)
 - These 2 records were removed from the dataset for subsequent analysis
- The loan amount is highly varied, even after removal of 'default' outliers and has an extreme right skew
 - The *Amount* column was transformed using a log base 10 transformation to appropriate a Gaussian-like distribution for regression analysis

ORIGINAL AMOUNT DISTRIBUTION

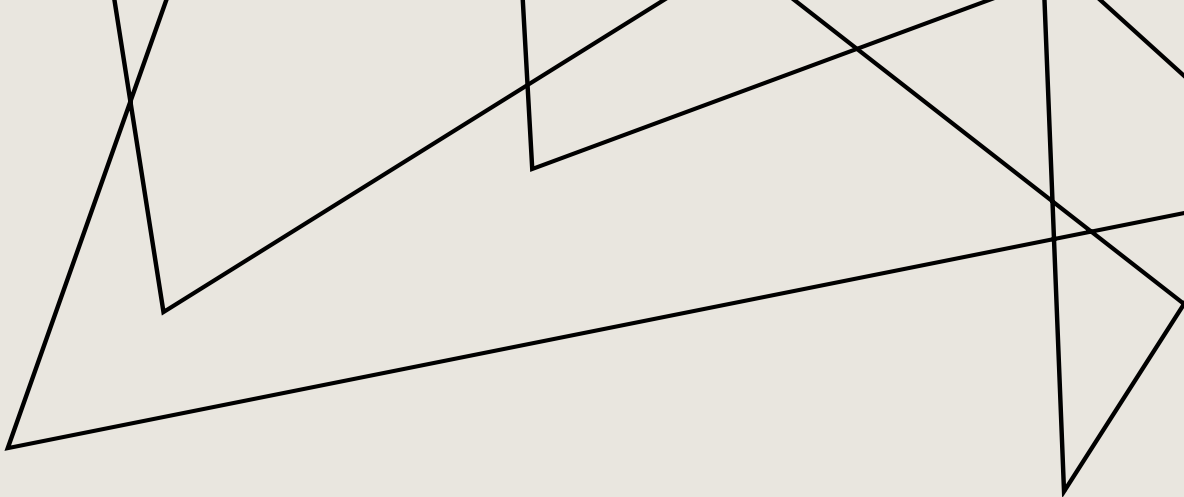


It is evident from that there is a high variance and a pronounced right skew in this data.

AMOUNT DISTRIBUTION POST-TRANSFORMATION

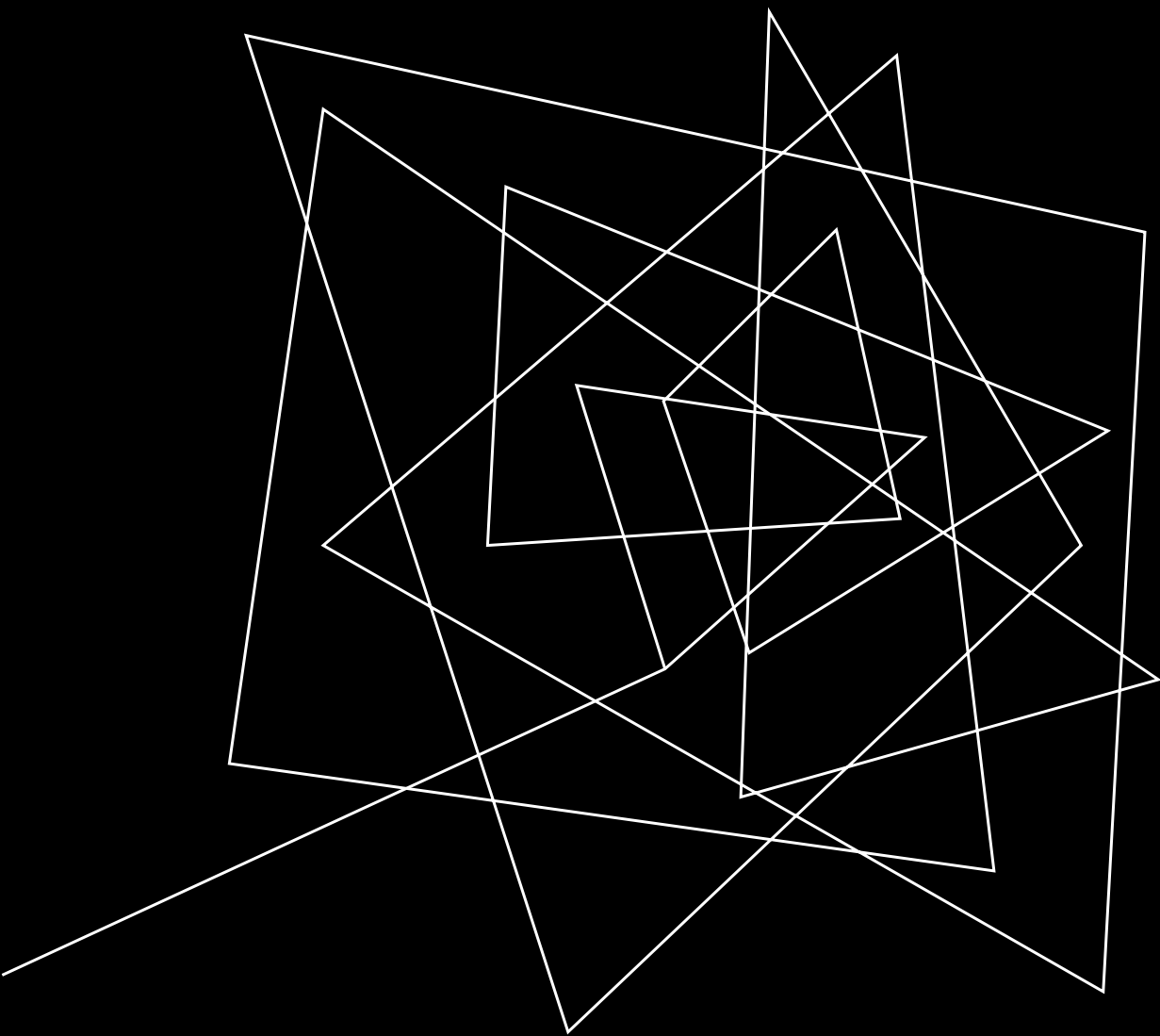


After using a log base 10 transformation, most of the skewness was adjusted. However, we do still see more weight with the left tail.



SUMMARY

- The *Amount* variable is highly varied and skewed
 - Transformation must occur to prepare the data for subsequent modeling or statistical analysis
- Using a log base 10 transformation, it can be approximated to a normal distribution
 - Comparing the empirical data (public_records) to theoretical cumulative density functions, the log10 data most closely follows the normal distribution
- It is of note to mention that there are a large portion of loans over \$1M
 - Focusing attention on high priority clients might improve customer satisfaction and potentially boost long-term revenue



QUESTION 3

REGARDING THE LOAN PERFORMANCE DATA, ARE THERE ANY RELATIONSHIPS BETWEEN PERFORMANCE METRICS?

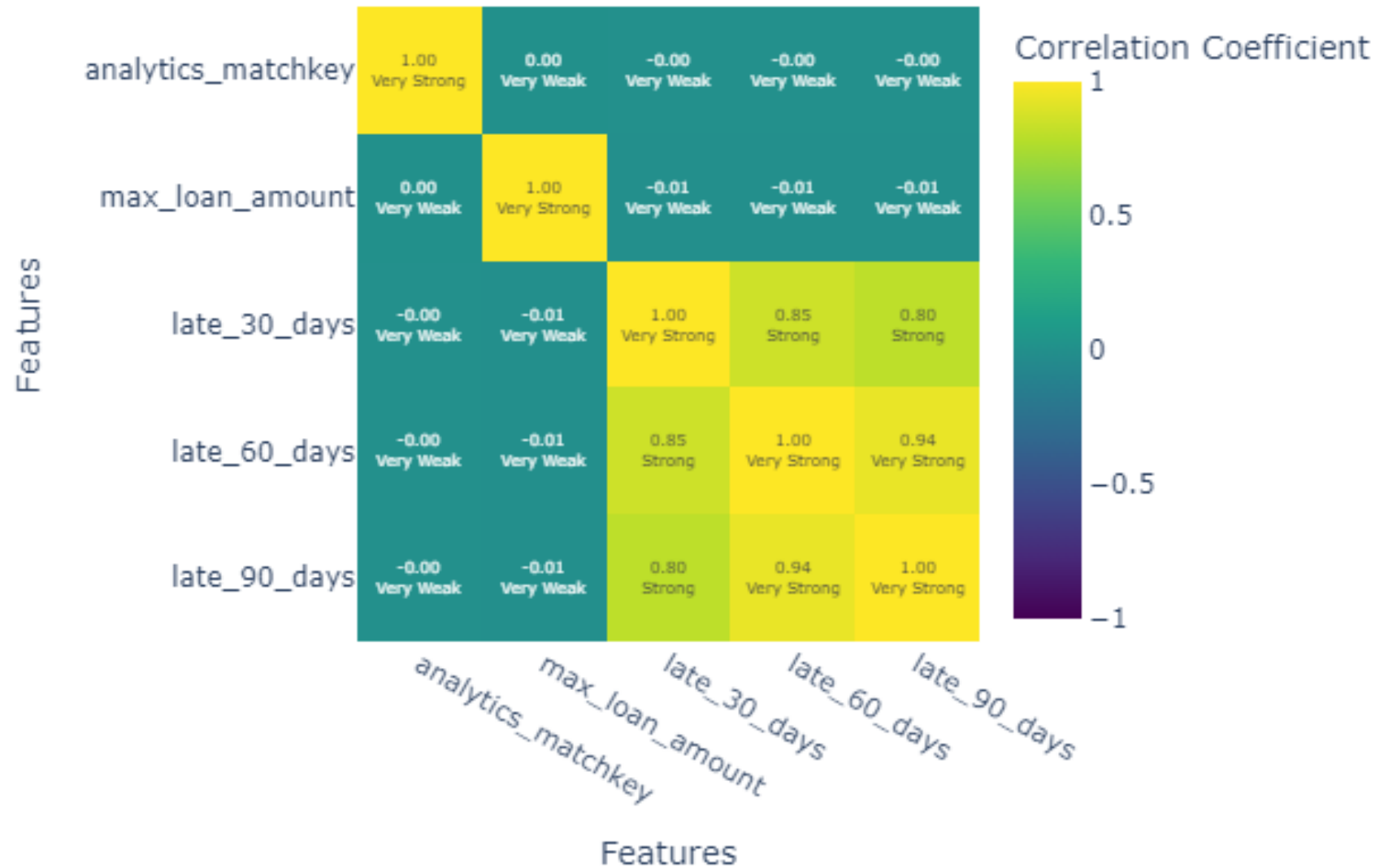


APPROACH

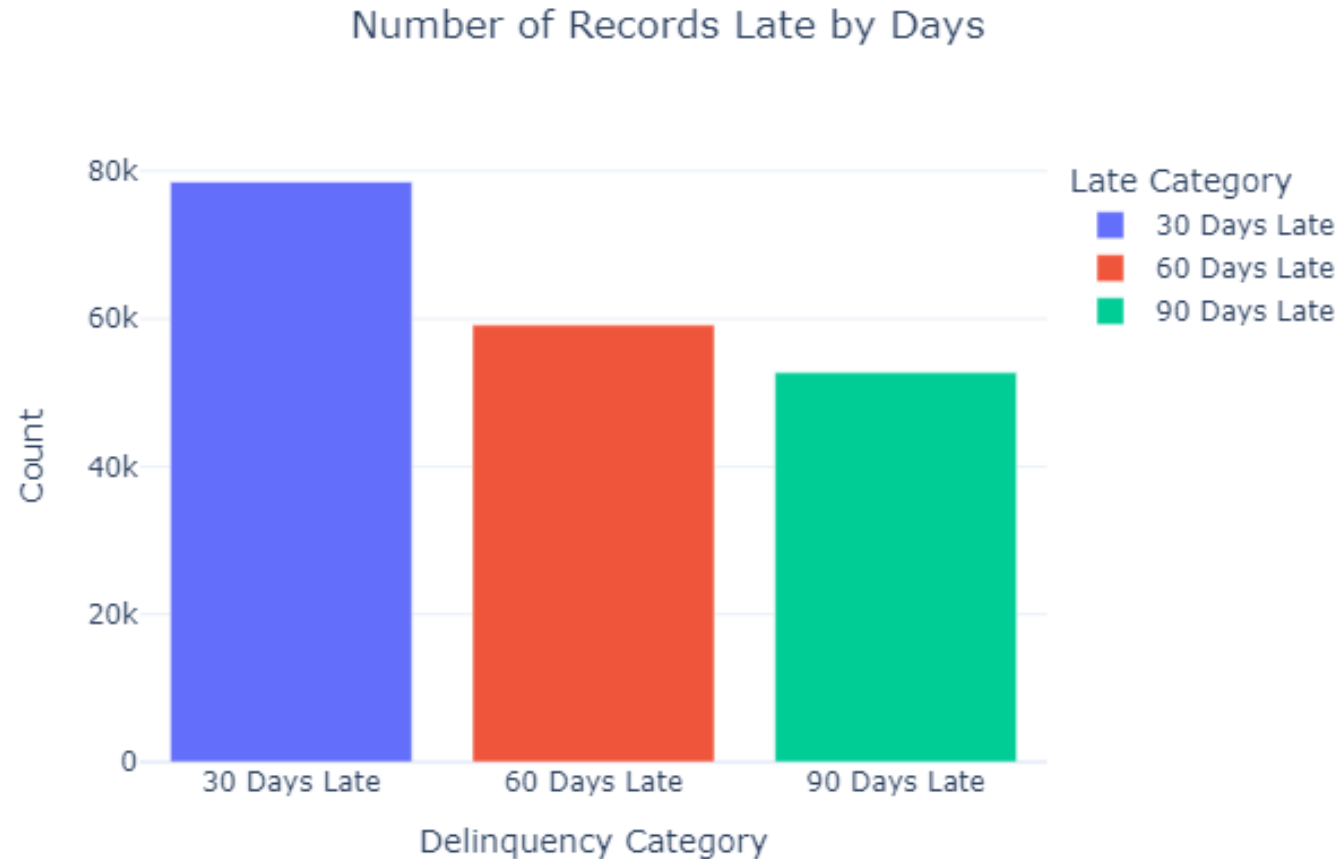
1. Examine the correlations between the variables in the data
2. Examine the distribution of delinquency
3. Conduct a Linear Regression
4. Conduct a Logistic Regression
 - Balancing class weights (delinquency vs. no delinquency)
 - Synthetic Minority Oversampling Technique (SMOTE)
5. Conduct a Random Forest analysis

CORRELATIONS

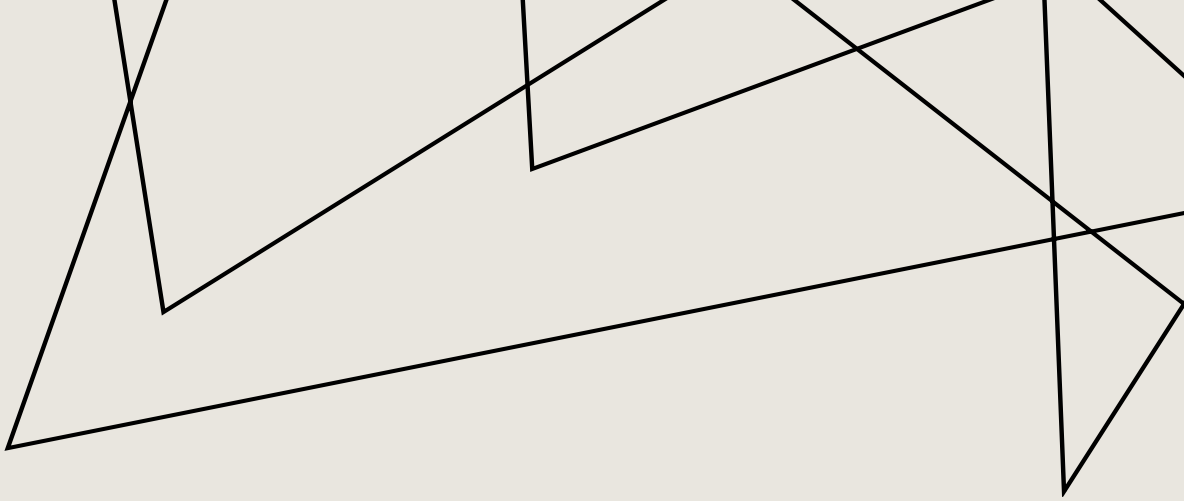
There does not seem to be a correlation between the maximum loan amount and any of the performance criteria. However, there are high correlations between being late on a loan and subsequent delinquency.



DISTRIBUTION OF DAYS OF DELINQUENCY

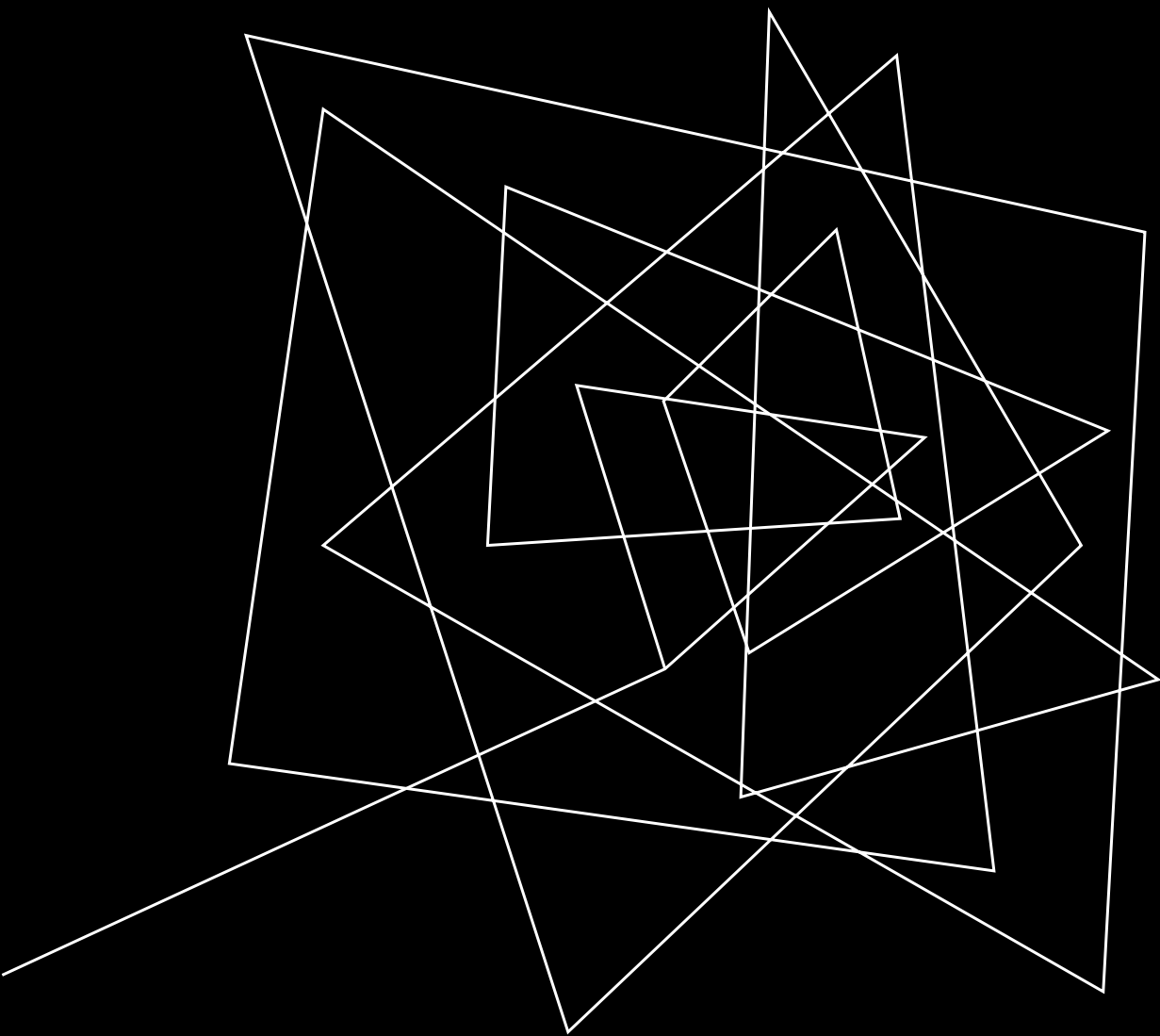


There are less loans that were delinquent more than 90 days than the other categories. However, there are still around 52K loans delinquent by at least 90 days, which is of significance.



SUMMARY

- Although there is a lower portion of loans that are delinquent by 90 days, which is a good sign, there are still around 52K in that category
 - Efforts could be focused on addressing these records to reduce the risk of possible default
- The output of regression analysis, before and after accounting for the discrepancy in class weights, showed that there is not a relationship that can be predicted between the maximum loan amount and the days of delinquency
- Even low loan amounts can perform poorly whereas high loan amounts can perform well
- More information is needed to accurately capture possible multivariate relationships



FURTHER EXPLORATION

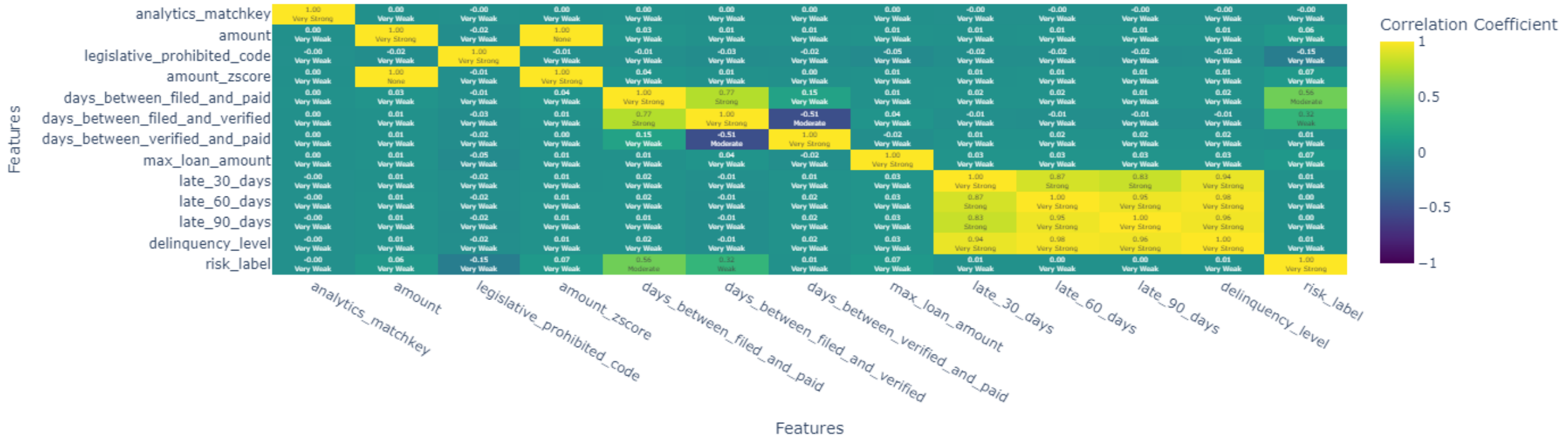
EXPLORE THE DATA IN THE WAY THAT YOU
WOULD TYPICALLY APPROACH IT.



APPROACH

1. Combine datasets to match performance to public record features
2. Examine correlations to explore if adding more features highlights potential relationships between variables
3. Analyze the dates associated with the record – filed date, verified date, and paid date – to see if there are any interesting trends
4. Create a risk profile from features previously analyzed to try to predict if potential clients will be at a high risk of default

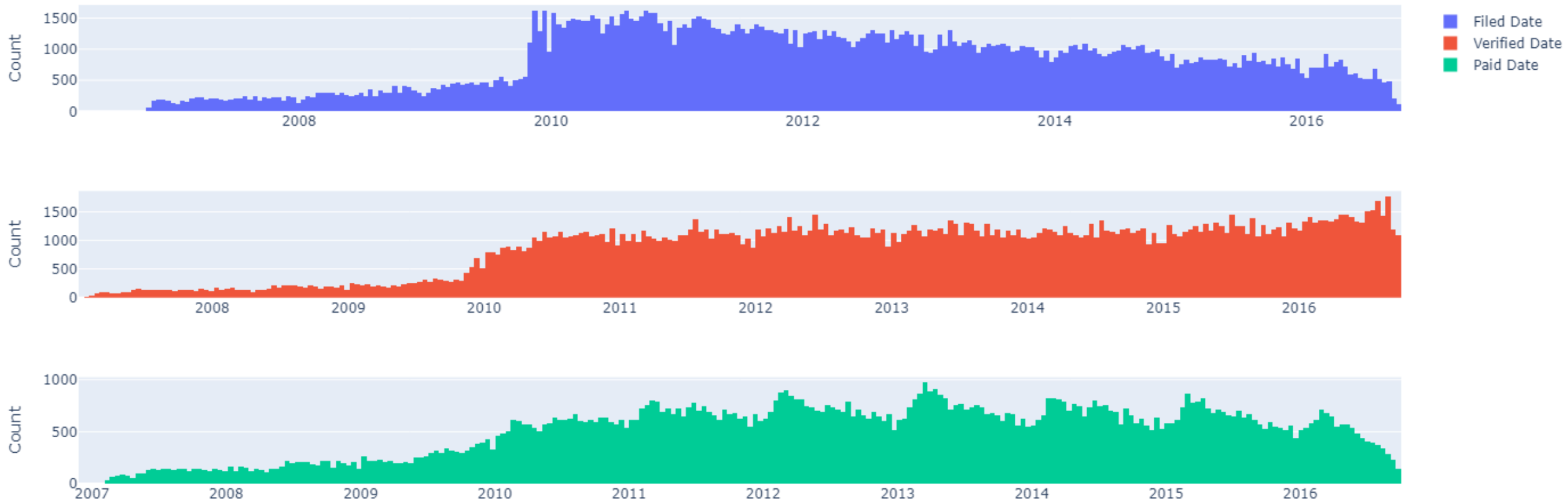
CORRELATION OF COMBINED FEATURES



Interestingly, there are no strong correlations when both datasets are combined, creating more feature interactions. Without additional data, it is difficult to make a distinct conclusion of what features might contribute to a high-risk profile.

DATE DISTRIBUTIONS

Distribution of Key Dates



Economic trends and cycles can be seen in this data. The recovery from the recession at the end of 2009 / beginning of 2010 led to more loans being filed and paid. Additionally, the highest count of paid loans is during Q1 while December showed the lowest count of loans paid.

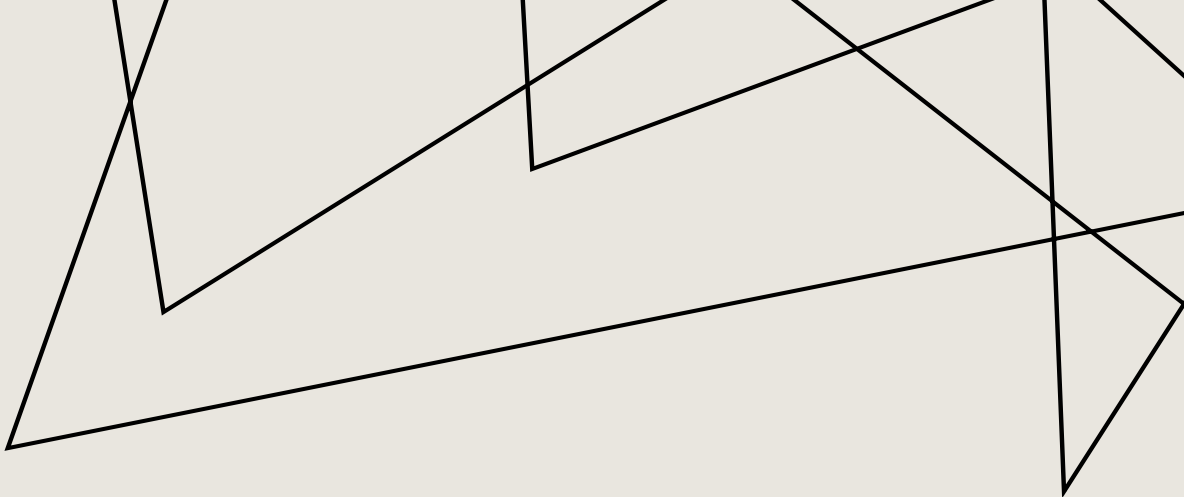
DEMONSTRATION OF RISK PROFILING

These two datasets do not show a strong propensity for being able to estimate the probability of default or to lend capabilities to building a risk profile due to the limited relational correlation between the features in the two datasets.

However, with additional data such as credit score, employment status, and income, a model could be constructed to predict potential risk scores for clients. If there were more features, I could assign a certain level of risk to each feature and, after validating the model with historical context and statistical analysis, assign weights to features shown to contribute to loan performance.

This was performed with the anomalies identified in Question 1, and a rough draft of how it could be implemented is given below.

	days_to_pay_back_loan	loan_amount	public_record_type_code	public_record_source_type_code	risk_percentage	risk_description
0	872.282947	171771.864061	FT	Other	45	This person is 45% likely to default on their loan.
1	1120.285672	186345.480792	Other	CC	80	This person is 80% likely to default on their loan.
2	468.335392	67946.974554	PF	CC	45	This person is 45% likely to default on their loan.
3	700.973393	141706.541202	FT	SU	70	This person is 70% likely to default on their loan.
4	869.781574	127792.444887	Other	CC	50	This person is 50% likely to default on their loan.



SUMMARY

- Analyzing time series data can shed light on economic and financial trends, lending tools to mitigate risk during periods of recession or economic hardship
 - These trends can also help identify yearly cyclical relationships between loan performance, providing a framework for operational interventions that could improve revenue and customer satisfaction
- When examining data to look for relationships and possible predictive power, it is important to understand the limitations of the dataset and the context in which models could be applied
 - Although the dataset given did not show a strong relationship between variables, a scaffold for a simulation model can be built that could account for more complexity were additional variables to be added into analysis

A series of white, overlapping geometric lines and polygons on a black background, located on the left side of the slide.

THANK YOU

Nikki Watson

702-461-6306

nikkiwat@umich.edu